

Lecture 12:

Ethical Considerations of SMDA

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

Recap: Causal Inference

What is causal inference?

- Causal inference: the process of determining *whether* (and *how much*) one variable **influences** another variable
 - Treatment (T): the cause we care about (e.g., exposure to a post, a platform ban)
 - Outcome (Y): the effect we measure (e.g., engagement, hate speech use)
- We want to estimate the causal effect of T on Y
 - Not just: are T and Y correlated?
 - But: would Y change if we changed T, holding everything else equal?

The fundamental question:

$$E[Y \mid \text{do}(T=1)] - E[Y \mid \text{do}(T=0)]$$

*"What would happen on average
if we forced T to be 1
vs. forced T to be 0?"*

Causal vs. descriptive questions

Same data, very different questions:

DESCRIPTIVE

What share of posts mention vaccines?

Are users in echo chambers more polarized?

Which subreddits have the most toxic comments?

How does engagement vary across topics?

CAUSAL

Does seeing pro-vaccine content increase intent to vaccinate?

Does joining an echo chamber make a user more polarized?

Did banning r/X reduce toxicity for its former members?

Does moral-emotional language cause higher engagement?

Watch the verbs: "is associated with" vs "causes" / "affects" / "changes"

Why is causal inference hard?

The fundamental problem of causal inference ([Holland, 1986](#)):

We can never observe the same unit both **treated** AND **untreated** at the same time.

FACTUAL (observed)

Alice was exposed to misinformation.

She shared 4 posts that day.

$Y(T=1) = 4$ ✓ **observed**

COUNTERFACTUAL (unobservable)

If Alice had NOT been exposed...

How many posts would she have shared?

$Y(T=0) = ?$ ✗ **never observed**

*Causal inference = the science of estimating the missing **counterfactual**.*

	Random assignment	No random assignment	
Random sampling	Causal inference, generalized to the whole population.	No causal inference, correlation statement generalized to the whole population.	Conclusions generalized to population.
No random sampling	Causal inference, only for the sample.	No causal inference, correlation statement only for the sample.	Conclusions not generalized to population.
	Causation	Correlation	

Mine Cetinkaya-Rundel's Statistics 101 Lecture:

https://www2.stat.duke.edu/courses/Fall12/sta101.001/resources/lecturettes/random_sample_assignment.pdf

When experiments aren't possible: observational data

- We often cannot (or should not) run experiments
 - Ethics: we can't randomly expose people to misinformation
 - Feasibility: we don't control the platform's algorithm
 - History: many questions are about events that already happened
- With observational data, **treatment is NOT randomly assigned**
 - Users self-select into being exposed to certain content
 - This selection is almost always confounded
- Two main strategies for observational causal inference:
 - Natural experiments — find a setting where treatment is "as good as random"
 - Adjustment-based methods — measure and condition on confounders (matching, regression, weighting)

~Weight Models: Matching

- Using matching as
 - a potential outcome prediction model



Causal Inference with Social Media Data

Experiments on social media platforms

- Researchers (sometimes) collaborate with platforms to run real experiments
 - Treatment: alter what some users see (feed content, notifications, ranking)
 - Outcome: measure resulting behavior (posting, engagement, sentiment)
- Example: [Kramer et al.](#) manipulated News Feeds for 600K+ Facebook users
 - Reduced positive (or negative) emotional content in some users' feeds
 - Found small but significant emotional contagion effects in users' own posts
- Famous and contentious; also raised major ethical concerns about consent

Experimental evidence of massive-scale emotional contagion through social networks

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Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved March 25, 2014 (received for review October 23, 2013)

Emotional states can be transferred to others via emotional contagion, leading people to experience the same emotions without their awareness. Emotional contagion is well established in laboratory experiments, with people transferring positive and negative emotions to others. Data from a large real-world social network, collected over a 20-y period suggests that longer-lasting moods (e.g., depression, happiness) can be transferred through networks [Fowler JH, Christakis NA (2008) *BMJ* 337:a2338], although the results are controversial. In an experiment with people who use Facebook, we test whether emotional contagion occurs outside of in-person interaction between individuals by reducing the amount of emotional content in the News Feed. When positive expressions were reduced, people produced fewer positive posts and more negative posts; when negative expressions were reduced, the opposite pattern occurred. These results indicate that emotions expressed by others on Facebook influence our own emotions, constituting experimental evidence for massive-scale contagion via social networks. This work also suggests that, in contrast to prevailing assumptions, in-person interaction and nonverbal cues are not strictly necessary for emotional contagion, and that the observation of others' positive experiences constitutes a positive experience for people.

computer-mediated communication | social media | big data

Emotional states can be transferred to others via emotional contagion, leading them to experience the same emotions as those around them. Emotional contagion is well established in laboratory experiments (1), in which people transfer positive and negative moods and emotions to others. Similarly, data from a large, real-world social network collected over a 20-y period suggests that longer-lasting moods (e.g., depression, happiness) can be transferred through networks as well (2, 3).

The interpretation of this network effect as contagion of mood has come under scrutiny due to the study's correlational nature, including concerns over misspecification of contextual variables or failure to account for shared experiences (4, 5), raising important questions regarding contagion processes in networks. An experimental approach can address this scrutiny directly; however, methods used in controlled experiments have been criticized for examining emotions after social interactions. Interacting with a happy person is pleasant (and an unhappy person, unpleasant). As such, contagion may result from experiencing an interaction rather than exposure to a partner's emotion. Prior studies have also failed to address whether nonverbal cues are

demonstrated that (i) emotional contagion occurs via text-based computer-mediated communication (7); (ii) contagion of psychological and physiological qualities has been suggested based on correlational data for social networks generally (7, 8); and (iii) people's emotional expressions on Facebook predict friends' emotional expressions, even days later (7) (although some shared experiences may in fact last several days). To date, however, there is no experimental evidence that emotions or moods are contagious in the absence of direct interaction between experimenter and target.

On Facebook, people frequently express emotions, which are later seen by their friends via Facebook's "News Feed" product (8). Because people's friends frequently produce much more content than one person can view, the News Feed filters posts, stories, and activities undertaken by friends. News Feed is the primary manner by which people see content that friends share. Which content is shown or omitted in the News Feed is determined via a ranking algorithm that Facebook continually develops and tests in the interest of showing viewers the content they will find most relevant and engaging. One such test is reported in this study: A test of whether posts with emotional content are more engaging.

The experiment manipulated the extent to which people ($N = 689,003$) were exposed to emotional expressions in their News Feed. This tested whether exposure to emotions led people to change their own posting behaviors, in particular whether exposure to emotional content led people to post content that was consistent with the exposure—thereby testing whether exposure to verbal affective expressions leads to similar verbal expressions, a form of emotional contagion. People who viewed Facebook in English were qualified for selective assignment to the experiment. Two parallel experiments were conducted for positive and negative emotion: One in which exposure to friends' positive emotional content in their News Feed was reduced, and one in which exposure to negative emotional content in their News Feed was reduced. In these conditions, when a person loaded their News Feed, posts that contained emotional content of the relevant significance, each emotional post had between a 10% and

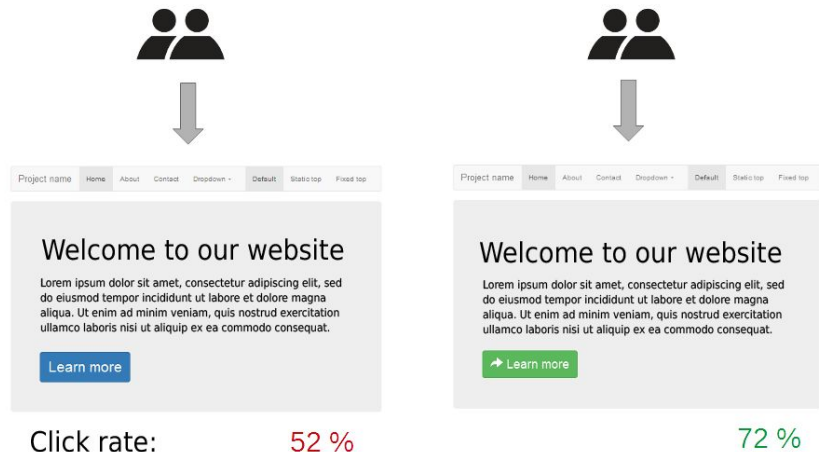
Significance

We show, via a massive ($N = 689,003$) experiment on Facebook, that emotional states can be transferred to others via emotional contagion, leading people to experience the same emotions without their awareness. We provide experimental evidence that emotional contagion occurs without direct interaction be-



Experiments on social media platforms: A/B tests

- A/B testing: industry term for a randomized controlled trial run on a live platform
 - Users are randomly split into group A (current experience) and group B (new variant)
 - The platform measures an outcome metric (clicks, time-on-site, retention) for each group
- Example of what gets A/B tested on social media
 - Feed ranking changes, notification timing, button placement, recommendation models
 - Even policy interventions: friction prompts, content warnings, "are you sure?" dialogs



https://en.wikipedia.org/wiki/A/B_testing

Experiments on social media platforms

- All social media experiments are not necessarily unethical or infeasible
- Create own experimental platform; similar to a lab study
 - High levels of control
 - Low external validity → might not generalize to real-world settings

Mahajan, K., Roy Choudhury, S., Levens, S., Gallicano, T., & Shaikh, S. (2021, March). [Community connect: A mock social media platform to study online behavior](#). In *Proceedings of the 14th ACM international conference on web search and data mining* (pp. 1073-1076).

Community Connect: A Mock Social Media Platform to Study Online Behavior

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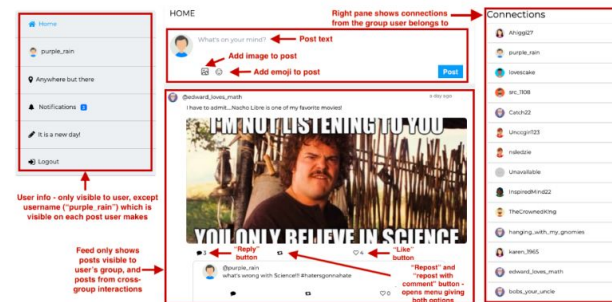


Figure 1: Interface of the Community Connect social networking platform and key features that are supported

ABSTRACT

We present Community Connect, a custom social media platform for conducting controlled experiments of human behavior. The key distinguishing factor of Community Connect is the ability to control the visibility of user posts based on the groups they belong to, allowing careful and controlled investigation into how information propagates through a social network. We release this platform as a resource to the broader community, to facilitate research on data collected through controlled experiments on social networks.

CCS CONCEPTS

ACM Reference Format:

Khyati Mahajan, Sourav Roy Choudhury, Sara Levens, Tiffany Gallicano, Samira Shaikh. 2021. Community Connect: A Mock Social Media Platform to Study Online Behavior. In *Proceedings of the Fourteenth ACM International Conference on Web Search and Data Mining (WSDM '21)*, March 8–12, 2021, Virtual Event, Israel. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3437963.3441698>

1 INTRODUCTION

Social media platforms, including Twitter, Instagram and Facebook, have become extremely popular for studying patterns of human

Experiments on social media platforms

- All social media experiments are not necessarily unethical or infeasible
- Create own experimental platform; similar to a lab study
 - High levels of control
 - Low external validity → might not generalize to real-world settings
- Alternative: **Field experiments**
 - Experiments carried out in real-world settings
 - High external validity, less control

Munger, K. (2017). [Tweetment effects on the tweeted: Experimentally reducing racist harassment](#). *Political Behavior*, 39(3), 629-649.

Polit Behav (2017) 39:629–649
DOI 10.1007/s11109-016-9373-5



ORIGINAL PAPER

Tweetment Effects on the Tweeted: Experimentally Reducing Racist Harassment

Kevin Munger¹

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© Springer Science+Business Media New York 2016

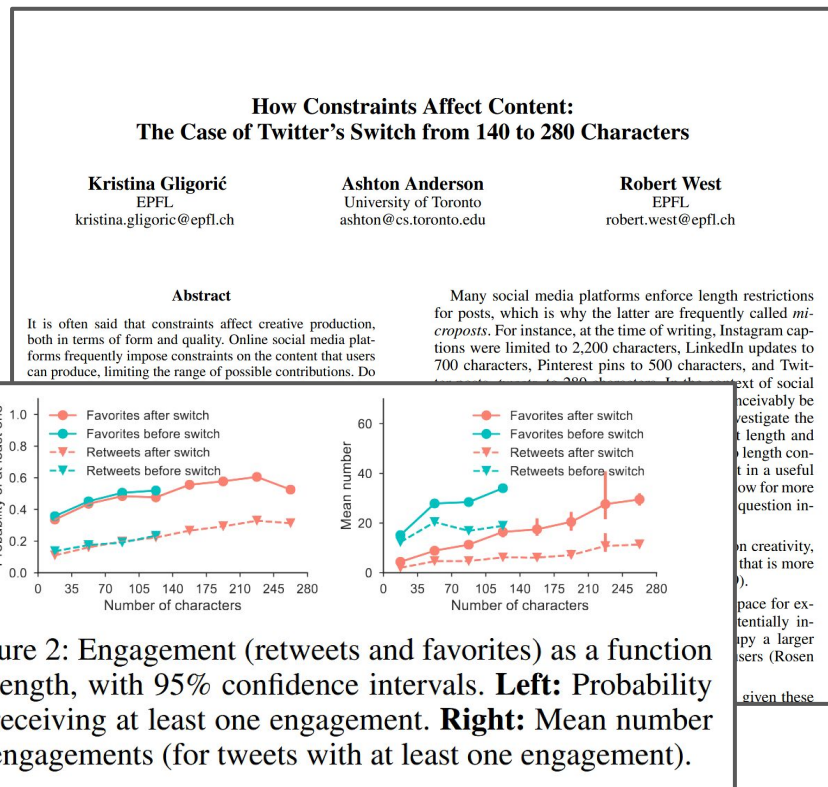
Abstract I conduct an experiment which examines the impact of group norm promotion and social sanctioning on racist online harassment. Racist online harassment de-mobilizes the minorities it targets, and the open, unopposed expression of racism in a public forum can legitimize racist viewpoints and prime ethnocentrism. I employ an intervention designed to reduce the use of anti-black racist slurs by white men on Twitter. I collect a sample of Twitter users who have harassed other users and use accounts I control (“bots”) to sanction the harassers. By varying the identity of the bots between in-group (white man) and out-group (black man) and by varying the number of Twitter followers each bot has, I find that subjects who were sanctioned by a high-follower white male significantly reduced their use of a racist slur. This paper extends findings from lab experiments to a naturalistic setting using an objective, behavioral outcome measure and a continuous 2-month data collection period. This represents an advance in the study of prejudiced behavior.

Keywords Online harassment · Social media · Randomized field experiment · Social identity

‘Natural’ Experiments on social media platforms

- Common sources on social media:
 - Sudden platform policy changes (e.g., bans, feed redesigns, deplatforming)
 - External shocks (news events, lockdowns, elections)
 - Staggered rollouts of features across users or regions
- Example: Do character limits on social media impact how people tweet?

Gligorić, Kristina, Ashton Anderson, and Robert West. "[How constraints affect content: The case of Twitter's switch from 140 to 280 characters.](#)" *Proceedings of the International AAAI Conference on Web and Social Media*.



Related: Event-centred studies

- Use a real-world event as the "treatment" — a natural experiment
 - Did something change after event X that we wouldn't expect otherwise?
- Common designs: Difference-in-differences (DiD) or Regression Discontinuity
- Key assumption: the event is plausibly **exogenous** to the outcome
 - Exogenous: outside the model, i.e., not associated with the treatment
 - Opposite of endogenous

Event-centred study: example

- RQ: Does deplatforming hate organizations actually reduce hate on the platform?
 - Treatment: Facebook removes leadership of designated hate organizations
 - Outcome: hate-content consumption + production by the audience
- Design: difference-in-differences across six staggered organization-level events
 - Treated: members of the hate organization's audience
 - Control: similar users not exposed to the disruption
 - Compare changes in hateful content before vs after the event
- Finding: deplatforming reduced consumption and production of hateful content on average
 - Some short-term backlash among the closest core members

Thomas, D. R., & Wahedi, L. A. (2023). [Disrupting hate: The effect of deplatforming hate organizations on their online audience](#). *Proceedings of the National Academy of Sciences*, 120(24), e2214080120.

PNAS

RESEARCH ARTICLE | POLITICAL SCIENCES

OPEN ACCESS



Disrupting hate: The effect of deplatforming hate organizations on their online audience

Daniel Robert Thomas* and Laila A. Wahedi*

Edited by Timothy Wilson, University of Virginia, Charlottesville, VA; received August 17, 2022; accepted January 20, 2023

How does removing the leadership of online hate organizations from online platforms change behavior in their target audience? We study the effects of six network disruptions of designated and banned hate-based organizations on Facebook, in which known members of the organizations were removed from the platform, by examining the online engagements of the audience of the organization. Using a differences-in-differences approach, we show that on average the network disruptions reduced the consumption and production of hateful content, along with engagement within the network among periphery members. Members of the audience closest to the core members exhibit signs of backlash in the short term, but reduce their engagement within the network and with hateful content over time. The results suggest that strategies of targeted removals, such as leadership removal and network degradation efforts, can reduce the ability of hate organizations to successfully operate online.

hate speech | deplatforming | social networks | leadership removal

Hate-based, terrorist, and criminal organizations attempt to use online platforms to spread their ideology, recruit new members, and coordinate existing members (1). They attempt to leverage online networks because of their ability to spread information quickly and widely. To counter this, social media platforms reduce the ability of such organizations to organize on their platforms, often by removing the members, or “deplatforming” them.¹ To what extent does deplatforming successfully mitigate harms? To answer this question, we employ a differences-in-differences design, taking advantage of the staggered timing of six exogenous organization-level deplatforming events, and granular daily user-level observations.² We show a strong overall negative effect: Depatforming reduces hate on the platform.

We study the effects of “strategic network disruptions” (SNDs), a method of deplatforming in which identifiable core members of a hate-based organization are removed from the platform all at once, eliminating the online leadership of the organization. The goal of this approach is to disrupt the operation of the organization by removing key actors at the same time in order to make it more difficult for them to rebuild their audiences and regroup. If the approach is successful, then the organization’s online target audience should be exposed to less radicalizing content and hate, and decrease their production of it. We study the effects of the disruptions on this target audience.

We find that disruptions create a healthier platform on average: Members of the audience of the hate-based organizations reduce their consumption and production of hateful content, and engage less with other audience members. However, these average results mask substantial heterogeneity between audience members who engaged most with the core prior to disruption, and those least engaged. While the least engaged subgroup reduces its engagement with hateful content and with other audience members following the disruptions, the most engaged subgroup exhibits signs of backlash, increasing its consumption and production of hateful content,³ and engaging more with other audience

Significance

How does removing the leadership of online hate organizations from online platforms change behavior in their target audience? We study the effects of six network disruptions of designated and banned hate-based organizations on Facebook, in which known members of the organizations were removed from the platform, by examining the online engagements of the audience of the organization. We show that on average, the network disruptions reduced the consumption and production of hateful content, along with engagement within the network among audience members. The results suggest that strategies of targeted removals, such as leadership removal and network degradation efforts, can reduce the ability of hate organizations to successfully operate online.

Author affiliations: *Meta, Menlo Park, CA 94025

Author contributions: D.R.T. and L.A.W. designed research; performed research; analyzed data; and wrote

¹Organizations that proclaim a violent mission, engage in violence, and represent ideologies that promote hate are

Matching example: Conspiracy Theorists on Reddit

- RQ: why people are drawn to conspiracy communities
- Platform: Reddit
- Match future conspiracists (FC) with non-conspiracists (CC)
 - Reddit joining time
 - contribution volumes across different time spans
 - subreddits they contribute to
- Compare social factors

Phadke, S., Samory, M., & Mitra, T. (2021). [What makes people join conspiracy communities? Role of social factors in conspiracy engagement.](#) *Proceedings of the ACM on Human-Computer Interaction*, 4(CSCW3), 1-30.

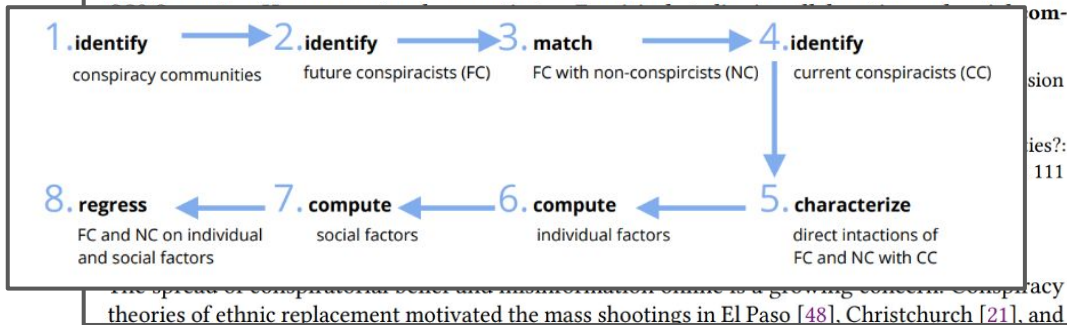
What Makes People Join Conspiracy Communities?: Role of Social Factors in Conspiracy Engagement

SHRUTI PHADKE, University of Washington, USA

MATTIA SAMORY, GESIS, Germany

TANUSHREE MITRA, University of Washington, USA

Widespread conspiracy theories, like those motivating anti-vaccination attitudes or climate change denial, propel collective action and bear society-wide consequences. Yet, empirical research has largely studied conspiracy theory adoption as an individual pursuit, rather than as a socially mediated process. What makes users join communities endorsing and spreading conspiracy theories? We leverage longitudinal data from 56 conspiracy communities on Reddit to compare individual and social factors determining which users join the communities. Using a quasi-experimental approach, we first identify 30K *future conspiracists*—(FC) and 30K matched *non-conspiracists*—(NC). We then provide empirical evidence of importance of social factors across six dimensions relative to the individual factors by analyzing 6 million Reddit comments and posts. Specifically in social factors, we find that dyadic interactions with members of the conspiracy communities and marginalization outside of the conspiracy communities, are the most important social precursors to conspiracy joining—even outperforming individual factor baselines. Our results offer quantitative backing to understand social processes and echo chamber effects in conspiratorial engagement, with important implications for democratic institutions and online communities.



Pitfalls to watch out for when doing Causal Inference

Carefully thinking about confounders

- List your suspected confounders BEFORE looking at the data
 - User-level: age, prior activity, network size, language
 - Content-level: topic, length, time-of-day
 - Platform-level: algorithmic exposure, recommendations
- You can only adjust for what you measure
 - Unmeasured confounders bias results
 - Within-study comparisons on Twitter found that even careful matching often fails to recover the experimental effect
- Be clear in limitations about potential unobserved confounders, e.g., views

Causal language: say what you actually mean

Match the strength of your verbs to the strength of your design.

DESCRIPTIVE / CORRELATIONAL

is associated with

correlates with

co-occurs with

predicts

users who post X also post Y

CAUSAL

causes

leads to

results in

increases / decreases

posting X makes users post more Y

Reviewers, journalists, and readers will (and should) hold you to the verbs you use. As will the evaluators of your course projects.

If your design is correlational, your title shouldn't say "X causes Y".

Today: Ethical Considerations when working with Social Media Data

What is Ethics?

- Ethics: “the discipline concerned with what is morally good and bad and morally right and wrong. The term is also applied to any system or theory of moral values or principles.” [[Britannica](#)]
- What should you take away from this lecture:
 - When analyzing social media data for research or beyond, how do you ensure that your work is ethical and does no harm?
 - How do you think about the ethical impacts of your projects?

Ethical Frameworks

- Systematic frameworks for thinking about moral decisions
- Five common ones:
 - **Deontology:** What is moral is what follows from absolute moral duties.
 - **Utilitarianism:** What is morally right is what generates the best outcome for the largest number of people.
 - **Virtue Ethics:** What is moral is what makes us the best person we could be.
 - **Rights-based Ethics:** What is moral is that which is in accord with everyone's rights.
 - **Care-based Ethics:** What is moral is that which promotes healthy relationships and the well-being of individuals and their interdependence.

[Penn State University: What are ethical frameworks?](#)

Ethics Review in the Universities

- You submit your research plan before executing it to an **Ethics Board** (also called Institutional review Boards or IRBs)
 - which methods and technical devices will be used,
 - how exactly data protection requirements will be fulfilled
 - are any invasive methods used
 - which risks for the human test subjects can be expected, and which precautions you will take to reduce risks or avoid/minimise problems that might arise during the experiment
- Highly recommended if you have human subjects in your study

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- Highly recommended if you have human subjects in your study
 - **But are observational studies with social media data also considered 'human subject' research?**

Do we need to talk about the ethical impacts of social media data collection and analysis?

Dual Use

- Dual use items: “refer to goods, software and technology that can be used for both civilian and military applications.”
 - More generally, harmful **reuse** of technology, science, and research
- Dual use dilemma: Ammonia, which was useful for agriculture, used for chemical warfare
- Are there dual uses of social media data analysis?

https://en.wikipedia.org/wiki/Dual-use_technology

Kaffee, Lucie-Aimée, et al. "[Thorny roses: Investigating the dual use dilemma in natural language processing.](#)"

Thorny Roses: Investigating the Dual Use Dilemma in Natural Language Processing

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Abstract

Dual use, the intentional, harmful reuse of technology and scientific artefacts, is an ill-defined problem within the contemporary context of Natural Language Processing (NLP). Large Language Models (LLMs) have advanced capabilities and become more accessible, increasing the risk of their intentional misuse by bad actors. To prevent such intentional misuse, it is necessary for NLP researchers to understand and document the potential harms of their research. Hence, we propose a specific definition of dual use for NLP researchers and practitioners. We propose a checklist for NLP researchers and practitioners to assess the potential for dual use of their research. The checklist is created based on research and practitioner

research artefacts. This is reflected in contemporary ethical review processes which emphasise the impacts of research on individual subjects, rather than on society as a whole.

Area	vulnerability	harms
NLP Applications	4.3	crime, oppression, manipulation
Ethics and NLP	4.1	ethics washing, surveillance, plagiarism, oppression
Psycholinguistics	4.0	cyber bullying, oppression
Generation	3.8	crime, cyber bullying, manipulation, oppression, ethics washing
Dialogue Systems	3.8	surveillance, crime
MT and Multilinguality	3.3	surveillance, crime
ML for NLP	3.2	military application, manipulation, oppression
Resources and Evaluation	3.1	ethics washing, surveillance, manipulation
Interpretability	3.0	ethics washing, manipulation
Information Extraction	2.8	surveillance, censorship
IR and Text Mining	2.7	surveillance

Case Study 1: Emotional Contagion on Facebook

[Kramer et al.](#) manipulated news feeds of Facebook users by reducing positive (or negative) emotional content

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Significance

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Case Study 2: AI-generated comments on r/changemyview

Researchers conducted a ‘field experiment’ by adding AI-generated comments on the subreddit r/changemyview to investigate AI-driven persuasion

Can AI Change Your View? Evidence from a Large-Scale Online Field Experiment

Keywords: Large Language Models, Online Experiments, Personalization, Persuasion, Reddit

Extended Abstract

Motivation. Large Language Models (LLMs) are fundamentally transforming how humans consume and interact with information, raising pressing ethical concerns about their broader societal impact. Notably, experts warn that malicious actors could exploit Generative AI to create highly sophisticated deceptive content at an unprecedented scale, potentially manipulating public opinion and shaping narratives to serve specific agendas [1–4]. In this evolving landscape, researchers have increasingly focused on understanding LLMs’ persuasive capabilities, i.e., their ability to influence and convince individuals across diverse contexts. Early studies on AI-driven persuasion have shown that LLMs can match human performance [5–9] or even surpass it [10–12], including when dealing with highly divisive sociopolitical issues. Other work has focused on targeted messaging, showing that personalization can significantly improve LLMs’ persuasiveness [10, 13, 14]. Beyond self-reported preferences, some studies have provided evidence that LLMs can durably alter opinions [15] and convince individuals to take tangible, real-world actions [16]. Despite these promising results, previous work faces fundamental limitations in ecological validity as it assesses LLMs’ persuasive capabilities within

https://regmedia.co.uk/2025/04/29/supplied_can_ai_change_your_view.pdf

How to reason about Ethics?

Ethics in the Digital Age

- Increased power and reach
 - All users of Facebook and r/changemyview
- Even for observational studies, there can be risks
- Concepts of informed consent, privacy, and human-subject research still in flux
 - Is it okay to use 'public' data for applications the data creator has not been informed about or consented to?
- Unintended effects
 - Other actors reuse openly available analysis code for malicious uses

Article

SM+S
social media + society

Social Media + Society
January-March 2018: 1–14
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DOI: 10.1177/2056305118763366
journals.sagepub.com/home/sms
SAGE

“Participant” Perceptions of Twitter Research Ethics

Casey Fiesler¹ and Nicholas Proferes²

Abstract

Social computing systems such as Twitter present new research sites that have provided billions of data points to researchers. However, the availability of public social media data has also presented ethical challenges. As the research community works to create ethical norms, we should be considering users' concerns as well. With this in mind, we report on an exploratory survey of Twitter users' perceptions of the use of tweets in research. Within our survey sample, few users were previously aware that their public tweets could be used by researchers, and the majority felt that researchers should not be able to use tweets without consent. However, we find that these attitudes are highly contextual, depending on factors such as how the research is conducted or disseminated, who is conducting it, and what the study is about. The findings of this study point to potential best practices for researchers conducting observation and analysis of public data.

Keywords

Twitter, Internet research ethics, social media, user studies

Introduction

In recent years, research ethics has become a topic of greater public scrutiny. This is particularly the case for research taking place on or using data from social computing systems (McNeal, 2014; Wood, 2014; Zimmer, 2010a). However, as Vitak, Shilton, and Ashktorab (2016) point out in a study of ethical practices and beliefs among online data researchers, there are not agreed upon norms or best practices in this

Traditional interventional human subjects research involves informed consent and direct interaction with participants who are aware they are being studied. Under certain regulatory conditions, research that falls outside of this scope may not be strongly regulated. For example, in the United States, the use of publicly available data (e.g., tweets) may not meet the criteria of research involving human subjects as per the Code of Federal Regulations (45 CFR 46.101). However,

Fiesler, Casey, and Nicholas Proferes. ““Participant” perceptions of Twitter research ethics.” *Social Media+ Society* 4.1 (2018)

Ethical Frameworks to Principles ([Salganik, “Bit by Bit”](#))

- **Respect for Persons:** treating people as autonomous and honoring their wishes.
- **Beneficence:** understanding and improving the risk/benefit profile of your study, and then deciding if it strikes the right balance.
- **Justice:** ensuring that the risks and benefits of research are distributed fairly.
- **Respect for Law and Public Interest:** include all relevant stakeholders, not just research participants

THE BELMONT REPORT

Office of the Secretary

Ethical Principles and Guidelines for the Protection of Human Subjects of Research

The National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research

April 18, 1979

AGENCY: Department of Health, Education, and Welfare.

ACTION: Notice of Report for Public Comment.

SUMMARY: On July 12, 1974, the National Research Act (Pub. L. 93-348) was signed into law, thereby creating the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. One of the charges to the Commission was to identify the basic ethical principles that should underlie the conduct of biomedical and behavioral research involving human subjects and to develop guidelines which should be followed to assure that such research is conducted in accordance with those principles. In carrying out the above, the Commission was directed to consider: (i) the boundaries between biomedical and behavioral research and the accepted and routine practice of medicine, (ii) the role of assessment of risk-benefit criteria in the determination of the appropriateness of research involving human subjects, (iii) appropriate guidelines for the selection of human subjects for participation in such research and (iv) the nature and definition of informed consent in various research settings.

[The Belmont Report](#)

Ethical Principles to Questions

- **Respect for Persons:** treating people as autonomous and honoring their wishes.
- **Beneficence:** understanding and improving the risk/benefit profile of your study, and then deciding if it strikes the right balance.
- **Justice:** ensuring that the risks and benefits of research are distributed fairly.
- **Respect for Law and Public Interest:** include all relevant stakeholders, not just research participants

Do you treat all research subjects with respect and dignity?

How do you minimize risks and maximize benefits?

Which people benefit from your research? Does it make the powerful more powerful?

What is the impact on people beyond your research subjects, e.g., the general public?

Putting Ethics into Action: Responsible AI and Science

- For research, reproducibility is paramount
- But sometimes, conflict with privacy, especially with sensitive subjects
- Also conflicts with dual use — malicious usage
- Recommendations*
 - Share your code
 - Share your data

* after considering costs and benefits

NeurIPS 2021 Paper Checklist Guidelines

The NeurIPS Paper Checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. For each question in the checklist:

- You should answer yes, no, or n/a.
- You should reference the section(s) of the paper that provide support for your answer.
- You can also optionally write a short (1-2 sentence) justification of your answer.

You are encouraged to take a look at the checklist early on as it likely will

Reviewers will be asked to use the checklist as one of the factors in their decision. It is preferable to "no", it is perfectly acceptable to answer "no" provided a proper justification is given. In general, answering "no" or "n/a" is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgement and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material.

We provide guidance on how to answer each question below. You may additionally refer to our [blog post](#) introducing the checklist to learn more about its motivation and how it was created.

1. For all authors...

- (a) Do the main claim
 - Claims in the paper
 - The paper's contribution as long as it is clear
- (b) Have you read the paper
 - Please read the paper

‘Just What do You Think You’re Doing, Dave?’* A Checklist for Responsible Data Use in NLP

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Abstract

A key part of the NLP ethics movement is responsible use of data, but exactly what that means or how it can be best achieved remain unclear. This position paper discusses the core legal and ethical principles for collection and sharing of textual data, and the tensions between them. We propose a potential

(Mulligan et al., 2019; Xiang and Raji, 2019) and “bias” (Blodgett et al., 2020). The result is differing expectations and standards, which makes it harder for fellow researchers to understand the terms of (re-)use of new resources and models, and also complicates peer review.

Experts in biomedical data research have concluded that “truly informed consent ... requires

Rogers, Anna, Timothy Baldwin, and Kobi Leins. “[Just What Do You Think You’re Doing, Dave? A Checklist for Responsible Data Use in NLP](#).” *Findings of the Association for Computational Linguistics: EMNLP 2021*.
The AI community has already started to work on the latter (Mantelero, 2018; Mittelstadt, 2019). The AI community has already started to work on the latter (Mantelero, 2018; Mittelstadt, 2019).

contribute to the development of a consistent standard for data (re-)use, embraced across

Reproducible Research Pipeline

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

<https://book.the-turing-way.org/reproducible-research/overview/overview-definitions>

Why is reproducibility and replicability important?

- Cornerstone of good science: reproducibility of empirical results is an essential part of the scientific method
- Ensure trustworthiness, rigor, and transparency of your findings
- Selfish reasons:
 - Better for you communicate what you did
 - Prevent mistakes:
 - Science is **iterative** and **self-correcting** —> honest mistakes happen
 - Providing reproducible materials out means that others can critically analyze and correct mistakes

Reasons behind the reproducibility crisis

- Fraud
- Technical debt:
 - “There’s a deadline, I’ll do this later”
 - “It’s just exploratory right now...”
- Lack of awareness of good practices
- Lack of incentives



Thinking about evidence, and vice versa

HOME TABLE OF CONTENTS FEEDBACK POLICY SEMINAR

[109] Data Falsificada (Part 1): "Clusterfake"

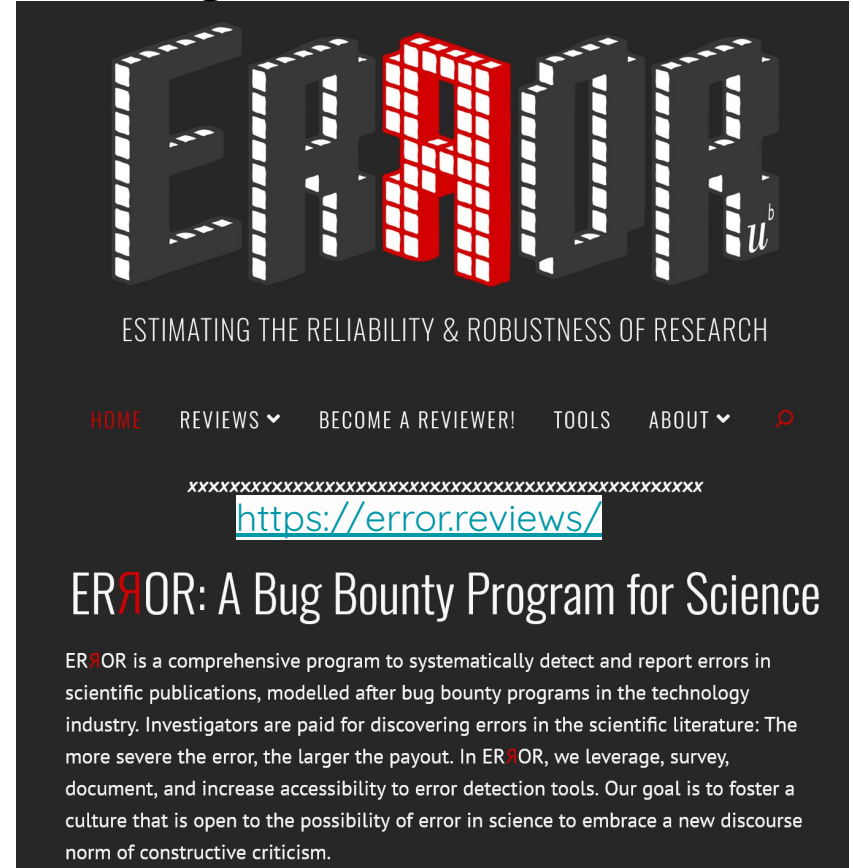
Posted on June 17, 2023 by Uri, Joe, & Leif

This is the introduction to a four-part series of posts detailing evidence of fraud in four academic papers co-authored by Harvard Business School Professor Francesca Gino.

In 2021, we and a team of anonymous researchers examined a number of studies co-authored by Gino, because we had concerns that they contained fraudulent data. We discovered evidence of fraud in papers spanning over a decade, including papers published quite recently (in 2020).

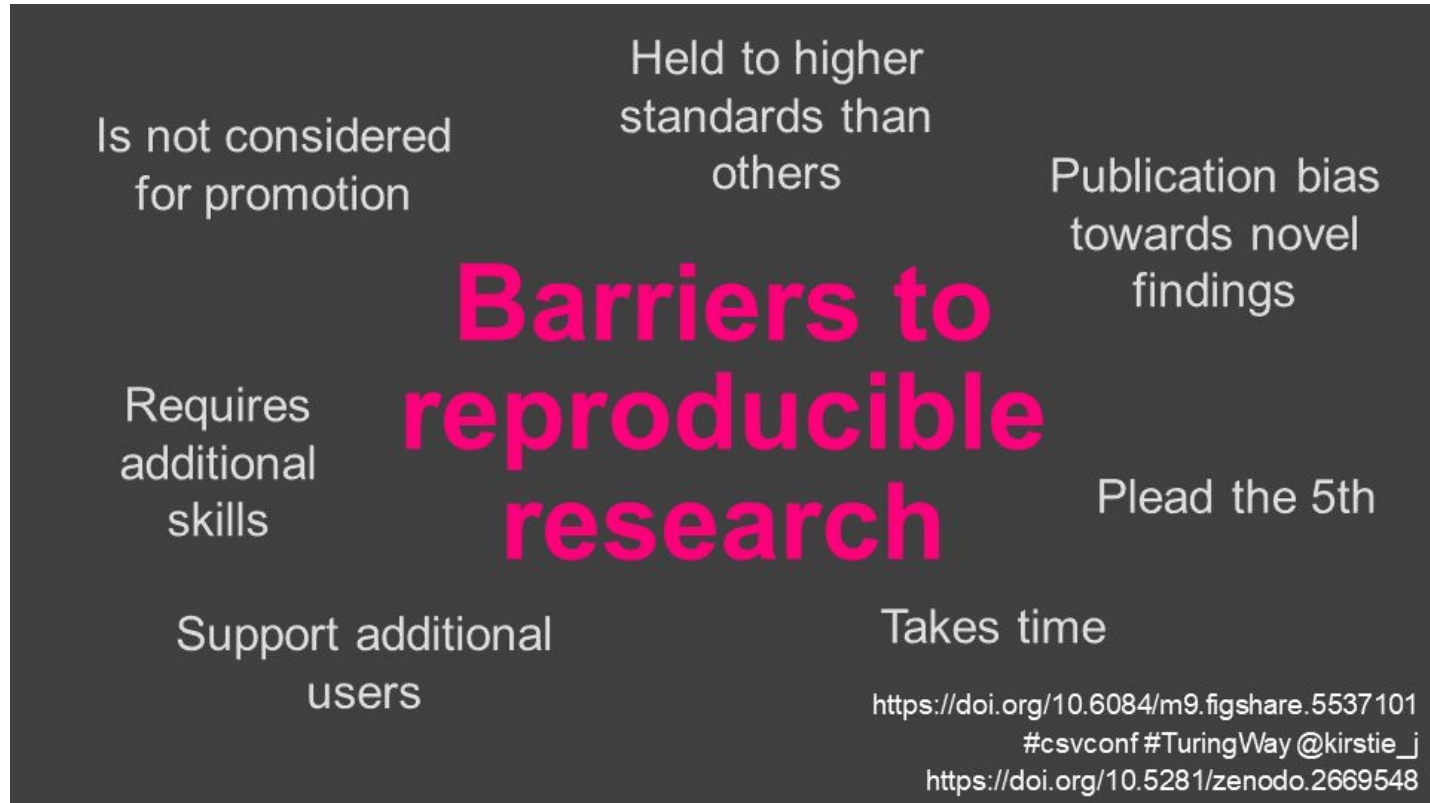
Reasons behind the reproducibility crisis

- Fraud
- Technical debt:
 - “There’s a deadline, I’ll do this later”
 - “It’s just exploratory right now...”
- Lack of awareness of good practices
- Lack of incentives



The screenshot shows the ERROR website. At the top, the word "ERROR" is displayed in large, 3D block letters. The letter "R" is red and has a grid pattern, while the others are grey. Below the title is the subtitle "ESTIMATING THE RELIABILITY & ROBUSTNESS OF RESEARCH". A navigation bar contains links: "HOME", "REVIEWS" (with a dropdown arrow), "BECOME A REVIEWER!", "TOOLS", "ABOUT" (with a dropdown arrow), and a magnifying glass icon. Below the navigation bar is a line of "x" characters, followed by a highlighted URL: <https://error.reviews/>. The main heading is "EROR: A Bug Bounty Program for Science". The text below describes the program: "EROR is a comprehensive program to systematically detect and report errors in scientific publications, modelled after bug bounty programs in the technology industry. Investigators are paid for discovering errors in the scientific literature: The more severe the error, the larger the payout. In EROR, we leverage, survey, document, and increase accessibility to error detection tools. Our goal is to foster a culture that is open to the possibility of error in science to embrace a new discourse norm of constructive criticism."

Common barriers to reproducibility



Common barriers to reproducibility

- Data
 - Inaccessible data
 - Poor or insufficient documentation
- Proprietary analysis tools, e.g., models or software
- Lack of reliability measures
 - **TIP:** always include error bars, confidence intervals
- Underpowered experiments
 - **TIP:** try to do a power analysis if you already have some exploratory results

Common barriers to reproducibility for CSS research

- Social media data sharing
 - Dataset decay
 - Tensions with welfare, privacy, and informed consent
 - Secondary data brokers providing little information about **data provenance**

Dataset	Lang	Size	Avail	Construct	Sharing	Hosting
Waseem2016 (Waseem and Hovy 2016)	EN	6,909	91/54%	HS	IDs	GitHub
WaseemHovy2016 (Waseem and Hovy 2016)	EN	16,791	64/50%	HS (sexism, racism)	IDs	GitHub
BenevolentSexism (Jha and Mamidi 2017)	EN	7,205	~/33%	HS (sexism)	IDs	GitHub
Davidson2017 (Davidson et al. 2017)	EN	25,296	•	HS & OL	Tweets	GitHub
Golbeck2017 (Golbeck et al. 2017)	EN	35,000	•	Harassment	Tweets on req	SH
Ross2017 (Ross et al. 2016)	GER	477	•	HS (immigrants)	Tweets	GitHub
Bohra2018 (Bohra et al. 2018)	EN,HI	4,067	69/69%	HS	IDs	GitHub
ElSherief2018 (ElSherief et al. 2018)	EN	28,498	~/32%	HS	IDs	GitHub
Founta2018 (Founta et al. 2018)	EN,HI	79,894	61/39%	HS	IDs / Tweets on req	SH / Zenodo
GermEval2018 (Wiegand, Siegel, and Ruppenhofer 2018)	EN,HI	8,541	•	OL	Tweets	GitHub
IberEval2018 (Fersini, Rosso, and Anzovino 2018)	EN	3,977	•	Sexism, misogyny	Tweets on req	SH
Rezvan2018 (Rezvan et al. 2018)	EN,HI	24,189	•	Harassment	Tweets on req	SH
Ribeiro2018 (Ribeiro et al. 2018)	EN	4,972	•	HS	Network(req)	GitHub / Kaggle
HASOC19 (Mandi et al. 2020)	EN,HI,GER	7,005	•	HS & OL	Tweets	SH
HatEval (Bassile et al. 2019)	EN,SP	13,000	•	HS	Tweets	SH
OLID (Zampieri et al. 2019)	EN	14,100	•	OL	Tweets	SH
Ousidhoum2019 (Ousidhoum et al. 2019)	EN,AR	5,647	•	HS	Tweets	GitHub
Toosi2019 (Toosi 2019)	EN	31,961	•	HS sentiment	Tweets	Kaggle
ALONE (Wijesiriwardene et al. 2020)	EN	688	•	HS/OL (Toxicity)	Tweets on req	SH
Gomez2020 (Gomez et al. 2020)	EN	149,823	48/44%	Multimodal HS	Tweets, IDs, Img	SH
MeTooMA (Gautam et al. 2020)	EN	9,973	78/78%	HS	IDs	Dataverse
CMSB (Samory et al. 2021)	EN	2,743	87/85%	Sexism	Tweets on req	Datorium
Covid2021 (Wich, Rätter, and Groh 2021)						
SWAD						

[The End of the Rehydration Era The Problem of Sharing Harmful Twitter Research Data](#)

GitHub

Common barriers to reproducibility for CSS research

[On the Challenges of Using Black-Box APIs for Toxicity Evaluation in Research](#)

- Social media data sharing
 - Dataset decay
 - Tensions with welfare, privacy, and informed consent
 - Secondary data brokers providing little information about **data provenance**
- Use of black-box services
 - Little information about service development (e.g., ChatGPT, Perspective API, Botometer)

On the Challenges of Using Black-Box APIs for Toxicity Evaluation in Research

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Abstract

Perception of toxicity evolves over time and often differs between geographies and cultural backgrounds. Similarly, black-box com-

Automatic toxicity detection tools, which often use machine learning algorithms to quickly analyze large amounts of data and identify patterns of toxic language, are a popular and cost-effective method

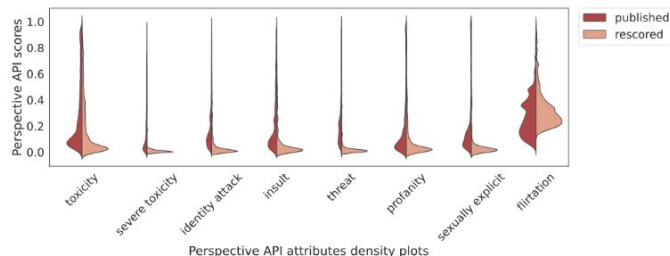


Figure 2: Rescored (Feb. 2023) and published (Sept. 2020) Perspective API attributes distributions from the RTP's prompts.

legal frameworks

research ethics

changing platform
access options / ToS

user expectations /
privacy

Tensions

platforms as
black boxes

data access
data sharing

interdisciplinarity

methods as
black boxes

publishing
practices

missing data

Different Conclusions When It Comes To Addressing Specific Challenges

E.g., in the context of **research ethics**:

- ? Big vs. small data
- ? Users as authors vs. users as research subjects
- ? Particularly vulnerable groups (e.g., activists) vs. professional / public accounts (e.g., politicians)
- ? Different practices in quoting from user accounts based on disciplinary requirements

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Weller, Katrin, and Katharina E. Kinder-Kurlanda. 2017. "[To Share or Not to Share?: Ethical Challenges in Sharing Social Media-based Research Data.](#)" In Internet Research Ethics for the Social Age, edited by Michael Zimmer, and Katharina E. Kinder-Kurlanda, 115-129. New York u.a.: Peter Lang.

Different Conclusions When It Comes To Addressing Specific Challenges

Or in the context of **data sharing**:

- ? balancing between following principles of good scientific practice and between respecting legal constraints
- ? Perceived ethical obligations *towards the scientific community*
- ? Not sharing data to protect users vs. sharing to include users

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Weller, Katrin, and Katharina E. Kinder-Kurlanda. 2017. "[To Share or Not to Share?: Ethical Challenges in Sharing Social Media-based Research Data.](#)" In Internet Research Ethics for the Social Age, edited by Michael Zimmer, and Katharina E. Kinder-Kurlanda, 115-129. New York u.a.: Peter Lang.

social media data aren't
„ordinary“ research data

Perceived ethical obligations *towards social media users*

“

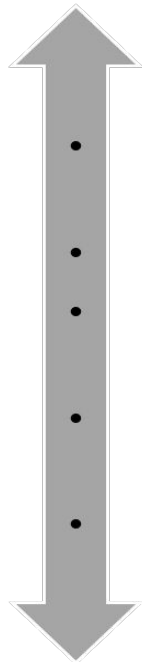
“It’s all public, it doesn’t belong to us, we don’t create the data, we don’t evoke it, I mean it’s natural. I don’t think you have the right to really keep other people from it, no.”

“

“We share datasets with everybody, actually. We don’t feel we own that.”

how much should I share?

Most reproducibility



What is being shared?

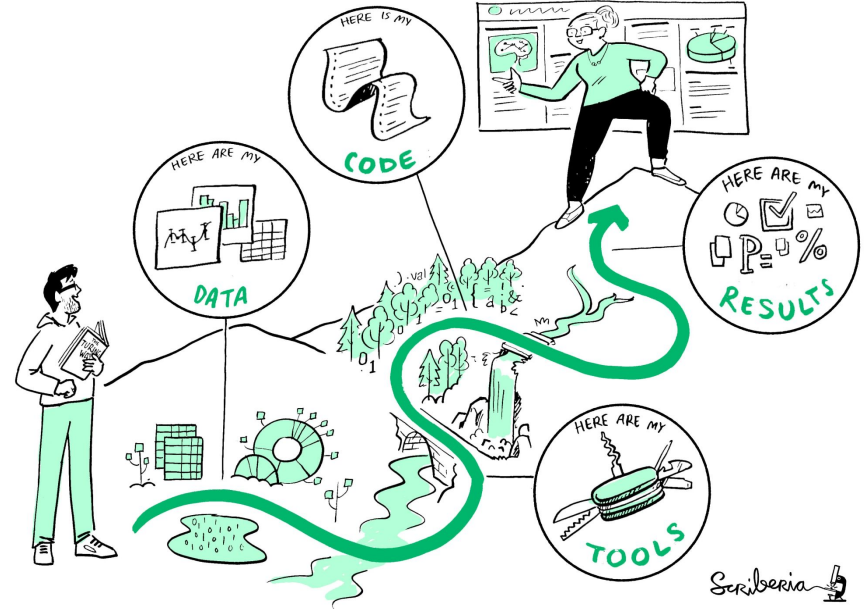
- whole dataset plus additional research information (e.g. scripts)
- whole dataset
- whole dataset, but without direct identifiers (pseudonymization)
- parts of the dataset removed (anonymization)
- changed dataset (e.g. only tweet IDs)

Most privacy

Weller, Katrin, and Katharina E. Kinder-Kurlanda. 2016. "[A manifesto for data sharing in social media research](#)." In Proceedings of the 8th ACM Conference on Web Science (WebSci '16), 166-172. New York: ACM.

How can you ensure your results are reproducible?

- Reproducible workflow from the very onset
 - Version control
 - Related: back up stuff regularly
 - Clear data management plan
 - Documentation
- When using social media data
 - Make sure the servers where your data is stored is secure



<https://book.the-turing-way.org/reproducible-research/reproducible-research>

How can you ensure your results are reproducible?

- When using black-box software
 - Save everything! For example, save the labels and not just the final analysis
 - Make a note of when the black-box software was used
- Training and testing models
 - Functions are your friends; make everything as modular as possible
 - Clear definition of inputs and outputs
 - Save the models
 - Save the model outputs!

Getting ready to publish your data and materials

- Make your data as accessible as possible
 - For sensitive cases, ensure privacy and anonymity
 - Consider uploading to data archives like Zenodo or Harvard Data Archive instead of github
 - Include a license: <https://choosealicense.com/licenses/>
 - For particularly sensitive cases, consider gated access
 - For newly created data resources (e.g., a newly annotated dataset) add a [data sheet](#) or [data statement](#)

Movie Review Polarity	Thumbs Up? Sentiment Classification using Machine Learning Techniques
Motivation	
For what purpose was the dataset created? Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description. The dataset was created to enable research on predicting sentiment polarity—i.e., given a piece of English text, predict whether it has a positive or negative affect—or stance—toward its topic. The dataset was created intentionally with that task in mind, focusing on movie reviews as a place where affect/sentiment is frequently expressed.¹	
Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)? The dataset was created by Bo Pang and Lillian Lee at Cornell University.	
Who funded the creation of the dataset? If there is an associated grant, please provide the name of the grantor and the grant name and number. Funding was provided from five distinct sources: the National Science Foundation, the Department of the Interior, the National Business Center, Cornell University, and the Sloan Foundation.	
Any other comments? None.	
Composition	
What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)? Are there multiple types of instances (e.g., movies, users, and ratings; people and interactions between them; nodes and edges)? Please provide a description. The instances are movie reviews extracted from newsgroup postings, together with a sentiment polarity rating for whether the text	
these are words that could be used to describe the emotions of john sayles' characters in his latest , limbo , but no , i use them to describe myself after sitting through his latest little exercise in indie egomania . i can forgive many things , but using some hackneyed , whacked-out , screwed-up * non * - ending on a movie is unforgivable . i walked a half-mile in the rain and sat through two hours of typical , plodding sayles melodrama to get cheated by a complete and total copout finale . does sayles think he's roger corman ?	

Figure 1. An example “negative polarity” instance, taken from the file neg/cv452.tok-18656.txt.

exception that no more than 40 posts by a single author were included (see “Collection Process” below). No tests were run to determine representativeness.

What data does each instance consist of? “Raw” data (e.g., unprocessed text or images) or features? In either case, please provide a description.

Each instance consists of the text associated with the review, with obvious ratings information removed from that text (some errors were found and later fixed). The text was down-cased and HTML tags were removed. Boilerplate newsgroup header/footer text was removed. Some additional unspecified automatic filtering was done. Each instance also has an associated target value: a positive (+1) or negative (-1) sentiment polarity rating based on the number of stars that that review gave (details on the mapping from number of stars to polarity is given below in “Data Preprocessing”).

Is there a label or target associated with each instance? If so, please provide a description.

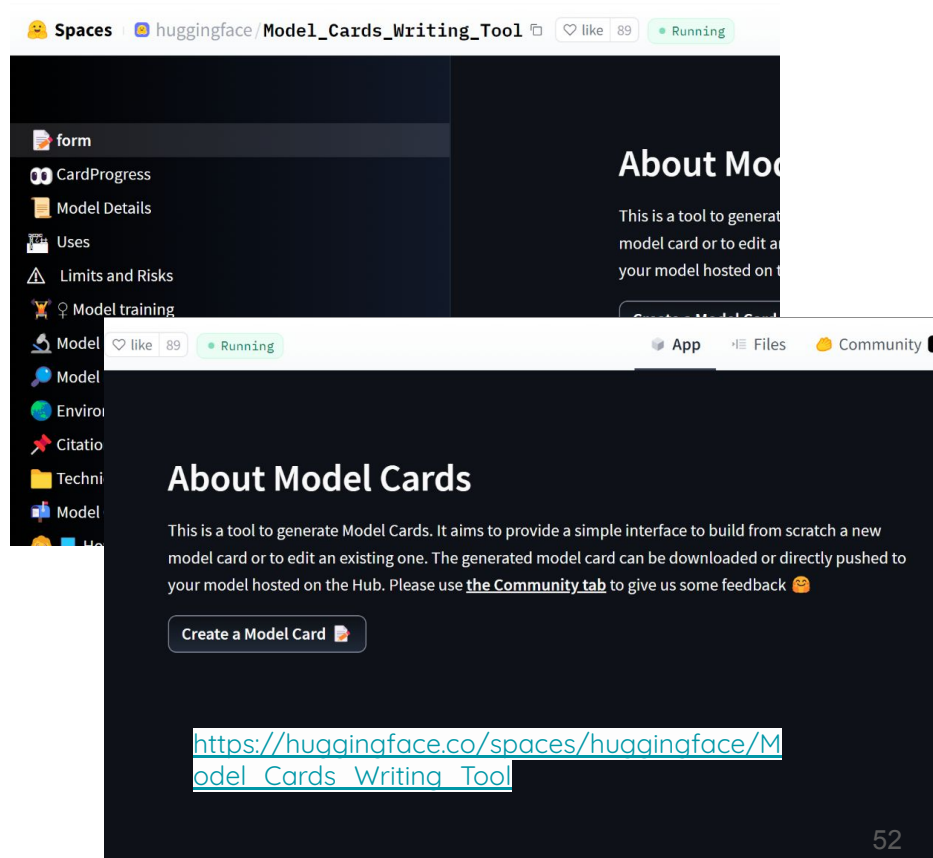
The label is the positive/negative sentiment polarity rating derived from the star rating, as described above.

Is any information missing from individual instances? If so, please

Datasheets for Datasets

Getting ready to publish your data and materials

- Respect data licensing agreements when using secondary dataset sources
 - Point to existing datasets rather than uploading a copy of it
 - Clearly describe how to access the data and how to preprocess it
 - Even better: upload the preprocessing script
- Code Documentation
 - Model cards



Class Activity

- With your project partners, discuss:
 - What are the benefits and harms of your project?
 - Could a malicious actor reuse your approach to cause harm?
 - Which **ethical frameworks and principles** do you use to justify your assessment?
 - Even if you do not have human subjects in your project, think about other stakeholder especially the broader public
 - How would you make your work reproducible?
- **Respect for Persons:** treating people as autonomous and honoring their wishes.
- **Beneficence:** understanding and improving the risk/benefit profile of your study, and then deciding if it strikes the right balance.
- **Justice:** ensuring that the risks and benefits of research are distributed fairly.
- **Respect for Law and Public Interest:** include all relevant stakeholders, not just research participants

Summary and Next Week

- This session we learned:
 - Ethical frameworks and principles
 - Reproducibility of social media data analysis
- Next week: **lecture on zoom**, link will be shared by next Monday on ILIAS!
 - Project presentation and report Dos and Don'ts
 - Project consultations
- No readings for next week
- Exercises next week will also be on project consultations, but in-person

Questions?

Additional Resources

- Ethics Board of Uni Mannheim:
<https://www.uni-mannheim.de/en/about/organization/bodies-and-committees/committees-and-councils/ethics-committee/#c160563>
- The Turing Way's Guide to Reproducible Research:
<https://book.the-turing-way.org/reproducible-research/reproducible-research>
- Course of effective research workflows: <https://www.brendanmichaelprice.com/workflow/>
focusing on many details like
 - Readable code
 - Collaborating well
- Schoch et al'2023 "[Computational Reproducibility In Computational Social Science](#)"